

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview

This document lists all the functional features implemented in the Credit Card Approval Prediction System. Each feature contributes to providing an intelligent, reliable, and user-friendly web application for predicting credit card approval, storing prediction history, and displaying analytics through a dashboard.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Applicant Information Form	Allows users to enter applicant details required for prediction.	HTML Form / Flask	Done	High
2	Input Validation	Validates user inputs before processing.	Flask Backend	Done	High
3	Data Preprocessing	Converts user input into model-compatible format.	Pandas / NumPy	Done	High
4	Credit Card Approval Prediction	Predicts whether the applicant is a Good Customer or Bad Customer using the trained ML model.	Random Forest Model	Done	Very High
5	Result Display	Displays prediction results on the web page.	HTML + Flask	Done	High
6	Responsive User Interface	Provides an attractive and responsive interface using HTML and CSS.	HTML / CSS	Done	Medium
7	Model Loading	Loads the trained model.pkl for prediction.	Pickle / Python	Done	High
8	Error Handling	Displays appropriate error messages for invalid or unexpected inputs.	Flask	Done	Medium

9	Confidence Score	Displays the confidence percentage of the machine learning prediction.	Scikit-learn / Flask	Done	High
10	Prediction History	Stores and displays previous applicant prediction records.	SQLite + Flask	Done	High
11	Dashboard & Analytics	Displays total predictions, approved customers, rejected customers, approval rate, and analytics.	Flask + SQLite	Done	Very High

Feature Summary:

Metric	Count / Value
Total Features Planned	11
Total Features Implemented	11
Core / Must-Have Features	7 (Input Form, Input Validation, Data Preprocessing, ML Prediction, Confidence Score, Result Display, Model Loading)
Additional / Nice-to-Have Features	7 (Input Form, Input Validation, Data Preprocessing, ML Prediction, Confidence Score, Result Display, Model Loading) • Responsive and attractive web interface. • Professional prediction result page. • Prediction confidence score. • SQLite database integration. • Prediction history management. • Dashboard with analytics and statistics. • Modular project structure for easy maintenance. • Easy deployment using Flask.
Features Tested & Verified	✓ Applicant input validation ✓ Machine learning model loading ✓ Credit card approval prediction ✓ Prediction confidence score generation ✓ SQLite database storage ✓ Prediction history retrieval ✓ Dashboard analytics ✓ Result display ✓ Responsive user interface ✓ Application execution without runtime errors

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Applicant Form, Result Page, Dashboard
2	Backend / Logic	5	Input Validation, Data Preprocessing, ML Prediction, Confidence Score, Error Handling
3	Database / Storage	2	SQLite Database, Prediction History
4	API / Integration	0	Not Required
5	Security / Authentication	0	Not implemented (not required for this project)
6	Analytics / Reporting	1	Dashboard Analytics